

VEDAANG CHOPRA

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MS CS (ML) at Georgia Tech — Advised by **Dr. Matthew Gombolay**, CORE Robotics Lab

Research: Coding agents with execution-grounded verification – Tool-use planning & multi-agent orchestration –
Evaluation harnesses that expose model failures

RESEARCH EXPERIENCE

Georgia Institute of Technology – CORE Robotics Lab @ Siemens

Atlanta, GA

Graduate Research Assistant — Advisor: Dr. Matthew Gombolay

Jan 2026 – Present

- Building a closed-loop coding-agent pipeline for parametric CAD code generation: a frozen LLM writes CadQuery programs, execution feedback (compile validity, geometric correctness via Chamfer/Hausdorff on STL/B-Rep) drives a LangGraph-orchestrated debug-and-repair loop – framing synthesis as inference-time search with verifiable geometric reward; 16-module system with a 202-test automated evaluation harness.
- Designing graders and failure diagnostics for generated code: AST/code-graph structural analysis, failure classification, and repair-trajectory quality scoring – turning messy qualitative behavior into concrete signals that steer iterative repair; VLM-based visual judgment complements programmatic metrics.

Georgia Institute of Technology – CS 8903 Special Problems

Atlanta, GA

Graduate Researcher

Jan 2025 – Present

- **SHASTRA** (Jan 2026–Present): Designing a trace-to-graph pipeline converting agent execution traces (GAIA, 94 sessions, 4,037 events, 442 graph nodes) into reusable task-composition graphs with dependency/control/conditional edges – compositional workflow reuse for new long-horizon tasks under cost/latency constraints.
- **ATHENA** (Jan–May 2025): Built a 5-agent supervisor-worker multi-agent system via LangGraph Command routing with reflection-based critique and dynamic plan modification; structured outputs validated with Pydantic; +0.18 BLEU-4 absolute over single-agent baseline on 100 reference videos; ablation showed removing the critique agent cut fact coverage 27%.
- **ARTEMIS** (Aug 2025–Present): Cost-aware VLM routing as dynamic tool selection – trained neural router picks the optimal model per request across 6+ backends served via vLLM with accuracy/cost/latency modes and SLA-aware load balancing.

INDUSTRY EXPERIENCE

Fortinet Technologies Inc. (AIOps R&D)

Bengaluru, India

Software Development Engineer I & II (ML/AI)

Feb 2021 – Jul 2025

- Led development of an agentic RAG diagnostics system from hackathon prototype to production deployment: LLM plans multi-step tool use over network telemetry, invokes APIs, and verifies hypotheses via structured function calling before concluding – autonomous root-cause analysis reducing mean resolution time 70%.
- Scaled OpenSearch ingestion pipelines 50→2,000 events/sec (40x) with Golang; deployed edge ML inference via ONNX Runtime (40% latency reduction); built SLA forecasting for 60+ classifiers across 4 categories.
- Patent (Filed): PCT/IN2022/058026 – Unsupervised distributional thresholding for wireless connectivity anomaly detection; reduced manual troubleshooting 75%.

TECHNICAL SKILLS

Agentic AI: Coding Agents, LangChain, LangGraph, Agentic RAG, Tool/Function Calling, Multi-Agent Collaboration, Long-Horizon Task Decomposition, Structured Generation (Pydantic), Workflow Composition

Evals & Verification: Automated Evaluation Harnesses, Grader Design (Execution-Grounded), LLM-as-a-Judge, Ablation Studies, Model Evaluation & Benchmarking, Failure Classification, Geometric/Programmatic Verifiers

AI / ML: PyTorch, Hugging Face Transformers, Deep Learning, Transformers, NLP, Computer Vision, Scikit-learn, NumPy, Pandas

LLMs / VLMs / Serving: vLLM, VLM Routing & Evaluation, Cross-Modal Retrieval, CLIP, SBERT, Inference Optimization

Systems: Python, Go, C/C++, SQL/PostgreSQL, Redis, OpenSearch/Elasticsearch, Docker, Linux, Git, FAISS, ONNX Runtime, FastAPI, Azure, W&B, TensorBoard

Reinforcement Learning: PPO, DQN, Reward Shaping, Curriculum Learning, Self-Play (academic projects)

EDUCATION

Georgia Institute of Technology

Atlanta, GA

M.S. in Computer Science (Machine Learning Specialization), **GPA: 4.0/4.0**

Aug 2024 – Dec 2026

- **Coursework:** Deep Learning, Deep RL, ML Security, Large & Vision Language Models, Agentic AI, Systems for AI

Maharaja Surajmal Institute of Technology

New Delhi, India

B.Tech. in Information Technology, **CGPA: 8.8/10.0**

Aug 2016 – Aug 2020